

Autodesk® Scaleform®

Scaleform 4.3

Scaleform 4.3

Matthew Doyle

2.10

: 2013 4 24

Copyright Notice

Autodesk® Scaleform® 4.3

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

Scaleform 4.3

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

1		1
2		2
2.1	Flash	2
2.2	SDK	2
2.2.1		3
2.2.2	SDK	4
2.2.3	Scaleform CLIK	4
2.2.4	Scaleform	5
2.2.5	Scaleform IME	5
2.2.6	Platform SDKs	5
2.2.7	.Net	6
2.2.8		6
2.2.9	Scaleform	8
2.3	Windows Scaleform SDK	13
2.3.1	Scaleform	13
2.4	Scaleform SDK for Mac	15
2.4.1		15
2.4.2		15
2.4.3	MacOS X(10.6 10.7)	15
2.4.4		16
2.5	Scaleform SDK	17
2.5.1	Scaleform SDK for Xbox 360	17
2.5.2	Scaleform SDK for PS3	19
2.5.3	Scaleform SDK for PS Vita	24
2.5.4	Scaleform SDK for Nintendo Wii	25
2.5.5	Scaleform SDK for Nintendo 3DS	27
2.6	Scaleform SDK	30
2.6.1	Android Scaleform SDK	30

2.6.2	iOS	Scaleform SDK.....	31
2.6.3			33
2.7	CS3	Scaleform	34
2.8	CS4, CS5	Scaleform	37
2.9	Scaleform		40
2.10			48
2.10.1			48
2.11			50
2.12			50
3			51
3.1	Flash.....		51
3.2			52
3.3			52
3.4			53
3.5			53
3.6			54
4			56
4.1			56
4.1.1	UI		56
4.1.2	UI		57
4.1.3	UI		57
4.2			57
4.3			58

1

Autodesk® Scaleform® 4.3
Scaleform 3.0

Scaleform 4.3

Scaleform Common Lightweight Interface Kit (CLIK™)

Scaleform

- Scaleform CLIK
- Scaleform CRI Movie™
- Scaleform (AMP™)
- Scaleform (IME)
-
-

: Scaleform 4.3 ActionScript 3.0 ,

2

Scaleform 4.3 SDK

Scaleform

Scaleform IME

<http://gameware.autodesk.com/scaleform/developer/?action=dl>

2.1 Flash

Scaleform 4.3	Adobe® Flash® Creative Suite® 3 (CS3)				Scaleform 4.3	
Flash Player 10.1	ActionScript™ (AS)		Scaleform 4.3		Flash	AS
	Scaleform 4.3		ActionScript 2.0		AS 3.0	
Flash CS3	CS4	AS 2.0	AS 3.0	Scaleform 4.3	Flash Player 10.1	AS
2.0	<i>Scaleform 4.3 – Flash</i>					

Adobe Creative Suite
Creative Suite
CS4 Flash Photoshop CS3
Illustrator

2.2 SDK

Windows ,Scaleform 4.3 SDK *C:\Program Files (x86)\Scaleform\GFX SDK 4.3*

Scaleform 4.3	DirectX® SDK	DirectX® SDK	SDK	Scaleform 4.3

Linux ,Scaleform SDK 4 , SDK
readme_make.txt readme_xx.txt

Scaleform 4.3	DirectX	Windows®
SDK		Windows

Scaleform->GFx SDK 4.3>Documentation
C:\Program Files (x86)\Scaleform\GFx SDK 4.3\Doc

2.2.1

Scaleform

1. (Full Source)
2. (Licensed Binary Libs)
3. (Evaluation Binary Libs)

Scaleform

SDK

Scaleform

Scaleform

Scaleform

()

(Lib)

GFXSDK

SDK

.tar.gz

.zip

Scaleform

Scaleform

(Scaleform Video)

make

Scaleform SDK

FxPlayer

PlayerSWFTToTexture

(

)

Visual Studio,

(Preprocessor Symbol)

GFX_ENABLE_VIDEO	Libgfxvideo
GFX_ENABLE_IME	Libgfxime

Scaleform

(Build)

(Shipping)

(Check)

(Release)

(Debug)

DebugOpt

2.2.1.1 Windows OS

tar.bz2 Windows , 7-zip tar.bz2

2.2.2 SDK

Scaleform 4.3 SDK Scaleform 4.3

Windows " " SDK → Scaleform → GfX SDK 4.3 → Scaleform
 SDK mac os scaleform_gfx_4.3_macos/Bin/Browser /
 Browser.app SDK

2.2.3 Scaleform CLIK

Scaleform 4.3 Scaleform (CLIK) UI
 CLIK SDK CLIK Flash
 CLIK CLIK *Getting Started*

1. Windows
 Start->Programs->Scaleform->GfX SDK 4.3->Documentation->Scaleform 4.3 – Getting Started with CLIK.pdf
2. Windows
C:/Program Files/Scaleform/GfX SDK 4.3/Doc/sf_4.3_getting_started_with_CLIK.pdf
3. SDK

, CLIK :
 Windows :
C:/Program Files/Scaleform/GfX SDK 4.3/Resources/AS2/CLIK/
C:/Program Files/Scaleform/GfX SDK 4.3/Resources/AS3/CLIK/
C:/Program Files/Scaleform/GfX SDK 4.3/Resources/Tools/
 :
scaleform_gfx_4.3_platform/Resources/AS2/CLIK/

scaleform_gfx_4.3_platform/Resources/AS3/CLIK/
scaleform_gfx_4.3_platform/Resources/Tools/

2.2.4 Scaleform

Scaleform CRI Scaleform
Scaleform 4.3 SDK Scaleform Video Flash ,
UI Flash , "PlayStation® 3" (PS3™) Xbox® 360 Wii™ Mac PC
Scaleform Video
Scaleform Scaleform 4.3 SDK Scaleform
Windows Scaleform->GFx SDK 4.3->Video
Scaleform

C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Tools/VideoEncoder/

C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Data/AS2/Video/

1. *C:/Program Files/Scaleform/GFx SDK 4.3/Doc/sf_4.3_getting_started_with_video.pdf*
2. Start->Programs->Scaleform->GFx SDK 4.3->Documentation->Scaleform 4.3 – Getting Started with Video.pdf
3. SDK Browser.

2.2.5 Scaleform IME

Scaleform (IME) Flash ,PC
Scaleform IME
Scaleform 4.3 SDK
C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Data/AS2/IME/
C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Data/AS3/IME/

2.2.6 Platform SDKs

Scaleform SDK, Scaleform SDK (Build History)

2.2.7 .Net

.NET 2.0 SP1 Scaleform CRI [Microsoft .NET Framework Version 2.0 SP1](#)

2.2.8

Scaleform Scaleform

3rdParty/

Apps/

Samples
Common
FxPlayer Flash
GFxPlayerTiny Scaleform

Bin/ ()
[Platform] (Scaleform Player)

Data/AS2/Samples	Sample Flash
FxPlayer	Scaleform Player
RenderTexture	RenderTexture
SWFToTexture	RenderTexture
ImageDelegate	ImageDelegate
Data/AS2/Video	Scaleform video
Data/AS2/IME	Scaleform IME
Data/AS2/Samples	Sample ActionScript2 Flash
Data/AS2/Kits	Sample data for ActionScript2 Kits
Data/AS3/Samples	Sample ActionScript3 Flash
Data/AS3/IME	Sample IME ActionScript 3
Data/AS3/Kits	Sample data for ActionScript3 Kits

Doc/

Include/ Scaleform 包括的文件 (C++ 便利头文)

Lib/ Scaleform

 [Platform]

 [Config]

Obj/

 [Platform]

Projects/ (make)

 Common make

 [Platform] make

Src/ Scaleform

 Render

 Sound

Scaleform #

Scaleform		#
Win32	Windows	SF_OS_WIN32
x64	Windows for x86_64	SF_OS_WIN32
*-linux	Linux for * processor	SF_OS_LINUX
MacOS	MacOS X	SF_OS_DARWIN, SF_C_OS_MAC

Xbox360	Xbox 360	SF_OS_XBOX360
---------	----------	---------------

	gcc
Msvc80	Visual Studio 8.0 (2005)
Msvc90	Visual Studio 9.0 (2008)
Msvc10	Visual Studio 10.0 (2010)
CW	Codewarrior
snc	SN

[Platform]

Win32/Msvc80

“ ”

MacOS

“ ”

2.2.9 Scaleform

make ,Scaleform Visual Studio make ,
 make Visual Studio IDE Projects() Solutions() 'Projects'

2.2.9.1 make

'make' , IDE(Visual Studio) Scaleform SDK
 readme_make.txt, make

- Linux
- MacOS
- Android™
- iPhone®
- PS3™
- Nintendo 3DS™
- Wii™
- Wii U™

make Scaleform , Linux , make cygwin,

1. <http://www.cygwin.com/>, setup.exe

2. “ ”

3. (c:/cygwin)

4. (c:/cygwin)

5. “ ”

6.

7. “ ” ‘Devel’ ‘binutils’ ‘make’

8.

, Windows "Cygwin Bash Shell", "ld" "make"

```
$ make -v
```

```
GNU Make 3.81
```

```
Copyright (C) 2006 Free Software Foundation, Inc.
```

```
...
```

```
This program built for i686-pc-cygwin
```

```
$ ld -v
```

```
GNU ld (GNU Binutils) 2.18.50.20080625
```

```
===
```

make utility Scaleform

make

1. Scaleform Makeconfig ()

a. export CELL_SDK := <path to cell sdk>

b. EPATH := \$(PATH)

c. export PATH = \$(CELL_SDK)/host-win32/ppu/bin:\$(CELL_SDK)/host-

 i. win32/Cg/bin:\$(CELL_SDK)/host-win32/bin:\$(EPATH)

2. Makeconfig unix msys
 /< >/... Cygwin /cygdrive/< >/... CELL_SDK
 C:\ps3\cell
 a. "export CELL_SDK := /cygdrive/c/ps3/cell".

3. , ,
 a. \$ make P=PS3
 b. \$ make P=PS3 C=Debug

make C=<config> "+" , RTTI
 , "make C=Release+NoRTTI" Make
 , Projects make </config>

2.2.9.2 'make'

make P=Android C=Debug
 make P=3DS C=Debug+NoRTTI

SNC **PS3**
 make P=PS3+snc

gcc **PS3**
 make P=PS3

SCU **SCU** **iOS**
 make P=iPhone/armv7 SCU=0

make P=wii C=Shipping+NoRTTI VERBOSE=1

x64 **Mac** **RTTI**
 make P=MacOS/x86_64 C=Release

iPhone **armv7**
 make P=iPhone/sim
 make P=iPhone/armv7 C=Release+NoRTTI

make P=MacOS C=Release+NoRTTI Lib/MacOS-x86_64/Release_NoRTTI/libgfx.a

4 (-j4)

make P=PS3 C=Shipping+NoRTTI -j4 Bin/PS3/FxPlayer_Shipping_NoRTTI.elf

2.2.9.3 SCU Builds

(Lumping) ,Scaleform (SCU)
(Unity build) SCU (SCU Builds)
Scaleform SDK ,
,
SCU , SCU , Scaleform

SCU _All.cpp , AS3 Obj Accessibility SCU
#include ,

Src/GFx/AS3/Obj/Accessibility/AS3_Obj_Accessibility_All.cpp

...
#include "AS3_Obj_Accessibility_ISearchableText.cpp"
#include "AS3_Obj_Accessibility_ISimpleTextSelection.cpp"
#include "AS3_Obj_Accessibility_Accessibility.cpp"
#include "AS3_Obj_Accessibility_AccessibilityImplementation.cpp"
#include "AS3_Obj_Accessibility_AccessibilityProperties.cpp"

SCU , SCU Visual Studio ,

SCU ,

SCU

- AS2_SCU
- AS3_SCU
- AIR_SCU

SCU

- AS2
- AS3
- AIR

, SCU


```
make ,SCU ,
'SCU=0' make ,
make P=PS3+snc C=Release+NoRTTI SCU=0
```

2.3 Windows Scaleform SDK

Windows® XP (32 bit, SP3), Vista (32/64 bit, SP2) Windows 7 (32/64 bit)
Visual Studio 2008 (9.0) Visual Studio 2010

Scaleform Scaleform

2.3.1 Scaleform

Scaleform

Scaleform

Scaleform SDK

\Program Files\Scaleform\GFx SDK 4.3\Lib\[Platform]\[MSVC Version]\

Debug	Scaleform	Debug libs
Runtime Library		Debug (/MTd)
Debug Information Format	C7	(/Z7)

DebugOpt	Scaleform
Runtime Library	(/MT)
Debug Information Format	C7 (/Z7)
Optimization	(/Ox)

Release	Scaleform
Runtime Library	(/MT)
Optimization	(/Ox)

Shipping	Scaleform	Shipping libs
Runtime Library	(/MT)	
Optimization	(/Ox)	

2.3.1.1 MSVC

Scaleform **SWFToTexture** **TextureInSWF** "Working
Directory" C:\Program Files (x86)\Scaleform\GFX SDK 4.3\Bin\Data

1. **Project**
2. **Properties**
3. **Configuration** All Configurations
4. **Debugging**
5. **Working Directory**

2.4 Scaleform SDK for Mac

Scaleform SDK Scaleform Cocoa GL MacOS

2.4.1

- Scaleform
- MacOS Snow Leopard (10.6)/Lion (10.7)
- Mac/iPhone/iPad Xcode(3.2.6)
- : PPC

2.4.2

Scaleform SDK for MacOS MacOS ,

- sf_4.3_eval_macos_i686.tar.bz2 sf_4.3_eval_macos_x86_64.tar.bz2

src

- sf_4.3_src_macos_i686.tar.bz2 sf_4.3_src_macos_x86_64.tar.bz2

() src

- sf_4.3_lib_macos_i686.tar.bz2 sf_4.3_lib_macos_x86_64.tar.bz2

Video

- Video
 - o sf_4.3_{eval/lib/src}_video_macos_i686.tar.bz2

2.4.3 MacOS X(10.6 10.7)

Scaleform MacOS SDK , terminal.app Scaleform

x86 x64 Scaleform()

\$ make P=MacOS

Bin/MacOS

```
CPU      Scaleform(      ):
$ make P=MacOS/i686
```

```
$ make P=MacOS/x86_64
```

```
,zlib
```

2.4.4

```
Bin/MacOS-i686  Bin/MacOS-x86_64                                (Finder)
```

```
$ Bin/MacOS-i686/FxPlayer.app/Contents/MacOS/FxPlayer <swf file>
```

```
AS2  AS3  ,      SWF      Dock      HUD      Menu
      Finder (Bin\Data\AS3\Kits\Menu or Bin\Data\AS3\Kits\HUD)
(      (Menu Kit)      $Bin/MacOS-i686/MenuDemo.app/Contents/MacOS/MenuDemo
```

2.5 *Scaleform SDK*

Scaleform

Scaleform 4.3 SDK

Scaleform

Xbox360®, Wii™ PS3™, PS Vita™ and 3DS™

2.5.1 Scaleform SDK for Xbox 360

Xbox 360

Xbox 360

Scaleform

2.5.1.1 /

Scaleform Xbox 360 SDK
Scaleform SDK

Scaleform 4.3 Xbox 360

Visual Studio

Visual Studio 2005 (Msvc80) Visual Studio 2008 (Msvc90)

Xbox 360 Development Kit Test Kit

Visual Studio

Debug

DebugOpt Shipping Release

Scaleform

Visual Studio

"Build"

"Build Solution"

Xbox Development Kit

GFX SWF

Xbox

Development Kit

Visual Studio

Xbox

"Set as StartUp Project"

Visual Studio

"Debug"

"Start Debugging"

F5

Xbox

PlayerTiny

Xbox

Test Kit

Xbox

Xbox

Xbox

A

Player

SWFToTexture

TextureInSWF

Xbox

PlayerTiny

(GFx_Xbox360, GFxExpat, Sound)

Xbox 360®

FxPlayer

(Gamepad)

- D Pad ()
- A (Enter)
- Y (Wireframe)
- X (Playback Info)
- B (Enable Fast Forward)
- RB LB
- " (Start) " / " (Pause/Play)
- Back(),

<back> ,FxPlayer

Flash

()

</back>

- A :Enter
- B :Escape
- Start : ~,
- LT :PgUp
- RT :PgDn
- LB:Home
- RB :End
- LTH :Insert
- RTH :Delete

FxPlayerXbox360.cpp

FXPLAYER_FILENAME

Flash (SWF)

#define FXPLAYER_FILENAME "Window.swf"

"Window.swf"

A

2.5.1.2

Scaleform (Xbox 360) Scaleform SDK
Xbox360 Scaleform

1. Scaleform
{GFX_SDK}/Projects/Xbox360/Msvc90/SDK/GFx 4.3 Xbox 360 3rd Party.sln (Release)
2. {GFX_SDK}/Projects/ Xbox360/Msvc90/SDK/GFx 4.3 Xbox 360 SDK.sln
3. Build Solution

RTTI () C++ Xbox360

2.5.2 Scaleform SDK for PS3

PS3 Scaleform SDK SDK “ readme ”

2.5.2.1

FMOD Cell SDK 250 GNU make version 3.81
Visual Studio

Visual Studio ProDG VSI PS3 SNC Visual Studio

libs libgfxrender_ps3.a Visual
Studio Visual Studio

libgcm *Lib/PS3/Shaders.a.*

PSGL *Lib/PS3/GLShaders.a.*

make

2.5.2.2

Lib/<platform>/libgfxrender_<renderer>.a

2.5.2.3 make

make Scaleform Linux make

1. Make Scaleform 2.2.9.1

2. Scaleform

\$ make P=PS3

To build using SNC:

\$ make P=PS3+snc

Lib Eval “ ” PS3 snc snc snc
make P=PS3+snc

3. SWF \$(CELL_SDK)/FxPlayer/flash.swf. SWF GFX
Scaleform Player L1 R1

4. " \$(CELL_SDK)/FxPlayer/sf_license.txt" (
) RTC PRX

cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);

5. ProDG Flash
 \$(CELL_SDK)/FxPlayer

6. SWFToTexture RenderTexture Bin/Data/AS2/Samples/SWFToTexture
 Bin/Data/AS2/Samples/RenderTexture SWF \$(CELL_SDK)/FxPlayer.

2.5.2.4 Visual Studio

Make	Visual Studio	Scaleform
PS3	Visual Studio	
	Visual Studio	
Scaleform	PS3	Visual Studio
Scaleform 4.3 PS3 SDK.sln	<i>Projects/PS3/Msvc90/SDK</i>	
Build	Build Solution	
		Scaleform 4.3 PS3 Demos.sln
<i>Projects/PS3/Msvc90/Samples</i>		Build
Solution		Build

2.5.2.5

"PS3"	Scaleform Player	,	"PS3"	Scaleform Player
,			"PS3"	Scaleform Player
"PS3"	Scaleform Player	,	"PS3"	Scaleform Player
			"PS3"	Scaleform Player
"PS3"	Scaleform Player	,	"PS3"	Scaleform Player

- D Pad ()
- (Cross) (Enter)
- (Triangle) (Wireframe)
- (Square) HUD
- (Circle)
- R1 L1
- L2 (Anti-Aliasing)
- R2 /
- " (Start) " / " (Pause/Play)
- Select(),

```

<select>      ,FxPlayer
ctor(PadKeyCommandMap  )      ,      FxPlayerAppBase
Flash

```

```

:"Window.swf"      D Pad      ,      (X)
"Window.swf"      /usr/local/cell/FxPlayer/flash.swf      FxPlayer      SWF/GFX
,      L1/L2

```

2.5.2.6 PS3

PS3 GCM GL

libgcm:

Src/Render/PS3/

Lib/PS3/Shaders.a

\$ make P=PS3 Lib/PS3/Shaders.a

PSGL:

Lib/PS3/Shaders.a

\$ make P=PS3 Lib/PS3/GLShaders.a

GLShaders.a SN "ppu-lv2-g++ -mno-sn-ld" GNU

libcgm PSGL Scaleform

libgcm

- PlayerGCM SWF
- PlayerTiny SWF
- SWFToTextureGCM SWF
- PlayerTextureInSWFGCM SWF

PSGL

- PlayerGL SWF

- PlayerTinyGL SWF
- SWFToTextureGL SWF
- TextureInSWFGL SWF

PSGL libgcm

PSGL *Apps/Samples/Common/OpenGLPS3App.**

SWFToTexture/TextureInSWF *Apps/Samples/Common/MathLib.**

```
#ifdef FXPLAYER_RENDER_GCM
    <gcm specific>
#else
    <psgl specific>
#endif
```

2.5.2.7

- Shipping
- Release
- Debug
- DebugOpt
- NoRTTI RTTI
- NoThreads Scaleform (SF_ENABLE_THREADS)

```
make C=<config> "+"
"make C=Release+NoThreads" Make
make
```

2.5.2.8

Scaleform PRX resc fs gcm

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RESC);
```

```
cellSysmoduleLoadModule(CELL_SYSMODULE_FS);
cellSysmoduleLoadModule(CELL_SYSMODULE_GCM);
```

```
ActionScript "Date" ( ) rtc
cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);
```

2.5.2.9

SF_ENABLE_THREADS

SF_ThreadsPthread.cpp

2.5.3 Scaleform SDK for PS Vita

2.5.3.1

Vita SDK

2.5.3.2

Scaleform SDK for PS Vita

PS Vita

- sf_4.3_psvita_vc90_eval.exe

src

- sf_4.3_psvita_vc90_src.exe

() src

- sf_4.3_psvita_vc90_lib.exe

2.5.3.3 /

Scaleform PS Vita SDK ,

Scaleform 4.3 PS Vita Demos Visual Studio

"Projects/PSVita/Msvc90/Demos/GFx 4.3 PSVita Demos.sln"

		FxPlayerMobile	FxPlayerTiny	(Release)	(Debug)	DebugOpt
(Shipping)		PS Vita	Scaleform		Scaleform	SDK
		PS Vita	Scaleform			

1. "Projects/PSVita/Msvc90/SDK/GFx 4.3 PSVita SDK.sln"
2. " (Build Solution)

make Vita , make Scaleform

2.2.9

2.5.3.4

		Demo	Source	
FxPlayerTiny		Scaleform		
FxPlayerMobile			HUD	

(SWF) Vita "File Serving Directory"() "FxPlayer"

SWF "flash.swf", FxPlayerMobile

2.5.3.5

, 2.10

2.5.4 Scaleform SDK for Nintendo Wii

2.5.4.1 /

Bin/wii Codewarrior ndrunk
Codewarrior (F5)

DVD Flash
RVL_SDK/dvddata/FxPlayer dvddata/FxPlayer/sf_license.txt.

```

Player          flash.swf (      SWF      GFx      )
PlayerTiny      Window.swf (      SWF      GFx      )
SWFToTexture    Bin/Data/AS2/Samples/SWFToTexture
                dvddata/FxPlayer
TextureInSWF    Bin/Data/AS2/Samples/RenderTexture
                dvddata/FxPlayer

```

```

FxPlayer
A
B      Escape
      /      HUD
      HUD
Home
-  +      Flash      (dvddata/FxPlayer      SWF      GFX      )
C
Z

<Z>      ,FxPlayer      ,      FxPlayerAppBase
ctor(PadKeyCommandMap      )      Flash

```

2.5.4.2

```

make files      Codewarrior
Scaleform

```

```

Shipping
Release
Debug
DebugOpt

```

```

+NoRTTI      RTTI
+Sdata0      (-sdata 0 -sdata2 0)
+NoThreads   Scaleform      (SF_ENABLE_THREADS;      )

```

```

        RTTI          +          Wii          +NoRTTI
        _ (          )

Codewarrior          Preferences          IDE Extras          “          ”
IDE          XML

Codewarrior          make

libgfx_video.a

C/C++          “ #define GFX_ENABLE_VIDEO ”

Scaleform          make

make          linux          Cygwin
PS3          2.3.2.3          “          make          Scaleform ”
Cygwin

SDK          Makeconfig          (          )          Wii SDK

WII_CWDIR := "/cygdrive/c/Program Files/Freescale/CW for Wii v1.0"
WII_SDKDIR := /cygdrive/c/RVL_SDK
WII_NDEVDIR := /cygdrive/c/NDEV

Makeconfig          unix
/cygdrive/<          >/...

cygwin shell          "make P=wii C=DebugOpt+NoRTTI"          DebugOpt+NoRTTI
C
Makeconfig          RVL_NDEV shell

```

2.5.5 Scaleform SDK for Nintendo 3DS

2.5.5.1

CTR SDK

2.5.5.2

Scaleform SDK for 3DS

3DS

- sf_4.3_eval_3ds.tar.bz2

src

- sf_4.3_src_3ds.tar.bz2

(
sf_4.3_lib_3ds.tar.bz2 src

2.5.5.3

Scaleform 3DS SDK , Cygwin Scaleform Scaleform():\$ make
P=3DS
Bin/3DS

make Scaleform , 2.2.9

2.5.5.4

FxPlayerMobile / FxPlayerTiny, Projects/3DS/Makefile STARTUP_FLASH
FxPlayer_CONTENT Flash , rom SWF
Obj/3DS/rom_FxPlayerMobile/FxPlayer cci
, CTR

2.5.5.5

, 2.10

2.6 **Scaleform SDK**

Scaleform

Scaleform

iOS

2.6.1 **Android Scaleform SDK**

2.6.1.1

Scaleform

- [Android SDK](#)
- [Android NDK](#)
- 适用于您的目标平台的适当 Android 平台 API

2.6.1.2

Android Scaleform SDK

Android

- sf_4.3_eval_android.tar.gz
src
- sf_4.3_src_android.tar.gz
- sf_4.3_lib_android.tar.gz

2.6.1.3 /

1. 在顶级 Scaleform 目录中创建一个 Makeconfig 文件（或者编辑那里提供的示例文件）。该文件应包含：
export JAVA_HOME := <path to JDK>
ANDROID_SDK := <path to SDK>
ANDROID_NDK := <path to NDK>
export PATH := \$(JAVA_HOME)/bin:\$(ANDROID_SDK)/tools:\$(ANDROID_SDK)/platform-tools

Makeconfig 中的路径必须是 unix 格式的路径，带有正斜杠，而且没有带冒号的驱动器号。将 `/<drive letter>/...` 用于 msys，而将 `/cygdrive/<drive letter>/...` 用于 Cygwin。例如，如果将 Android_SDK 安装到 C:\Android\SDK：

```
"export ANDROID_SDK := /cygdrive/c/Android/SDK".
```

2. 生成 Scaleform 库和示例：
\$ make P=Android
3. 将在 Bin/Android 中找到的适当 APK 文件复制到设备中。导航到设备上您放置 APK 的位置，并将其安装到设备。
4. 将一个 SWF 文件复制到 `/sdcard/flash.swf`。可以通过使用菜单键访问的弹出式 HUD 中的箭头来在 Scaleform 播放器中查看该目录中的其它 SWF 和 GfX 文件。
5. 如果这是一个评估试用版本，请将许可证密钥放置到 `/sdcard/sf_license.txt` 中。
6. 现在从您的应用程序启动程序运行演示版应用程序，就会加载 flash.swf copied to `"/sdcard"`。

2.6.1.4

Scaleform	Scaleform SDK	Android	Scaleform
Make			

2.6.2 iOS Scaleform SDK

2.6.2.1

Scaleform

- Mac OS X Snow Leopard
- Xcode 3.2.5, Xcode 4
- SDK

2.6.2.2

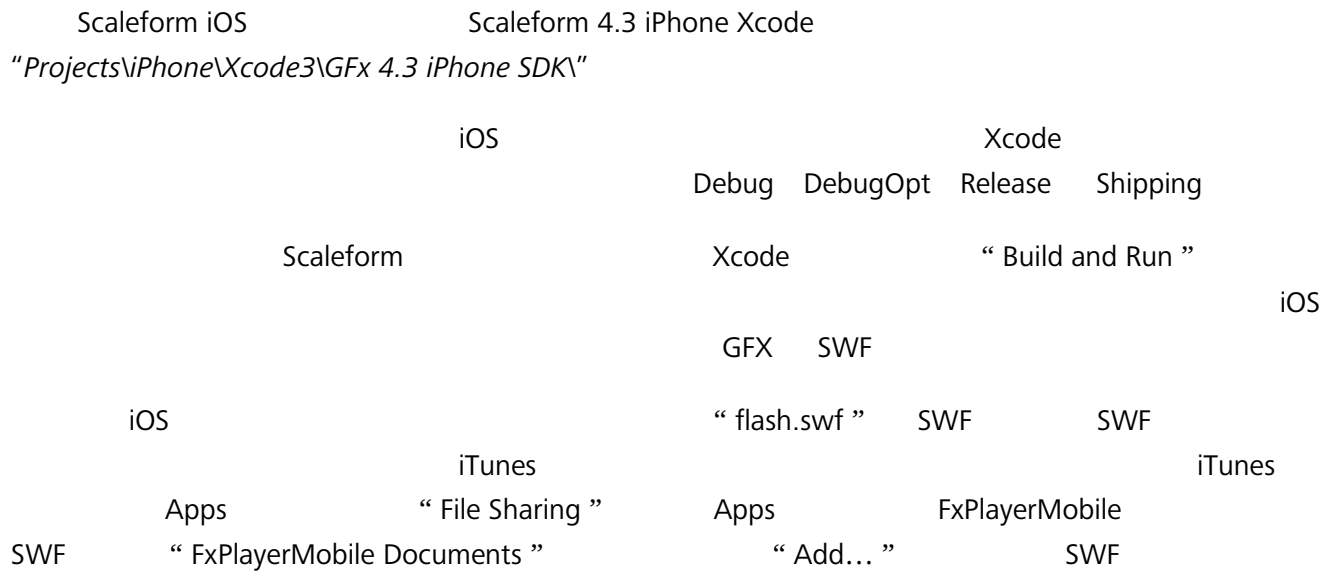
iOS	Scaleform SDK	iOS
-----	---------------	-----

- sf_4.3_eval_ios.tar.bz2

- src
- sf_4.3_src_ios.tar.bz2
- sf_4.3_lib_ios.tar.bz2

“ ”

2.6.2.3 /



2.6.2.4



2.6.3

Scaleform SDK

1. *onPause* *Scaleform*

, Gfx::Movie Advance Gfx::Movie Advance
NextCapture , , , ,

2. *Scaleform* *Scaleform*

/ (Advance) (Display) / ,

3. *HAL* *GL*

Render::GL::HAL::ResetCoNtext HAL / / ,
GL

4. *Scaleform*

, ResetCoNtext , Scaleform
(Advance)/ (Capture)/ (NextCapture) / (Display), /

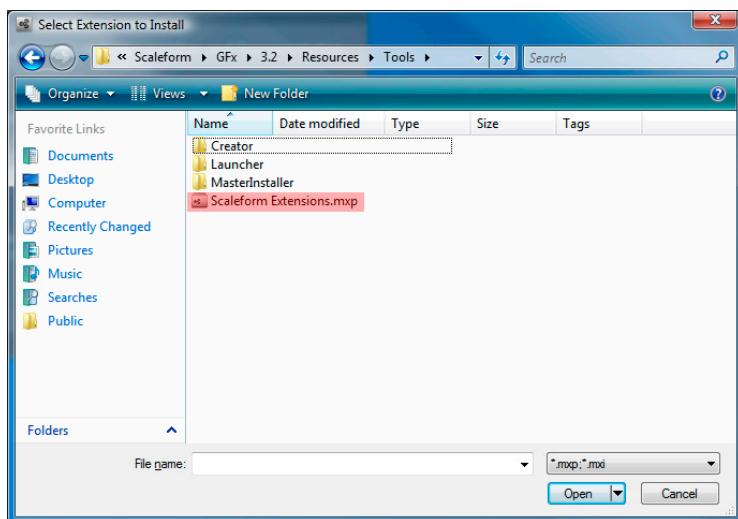
2.7 CS3 Scaleform

Scaleform

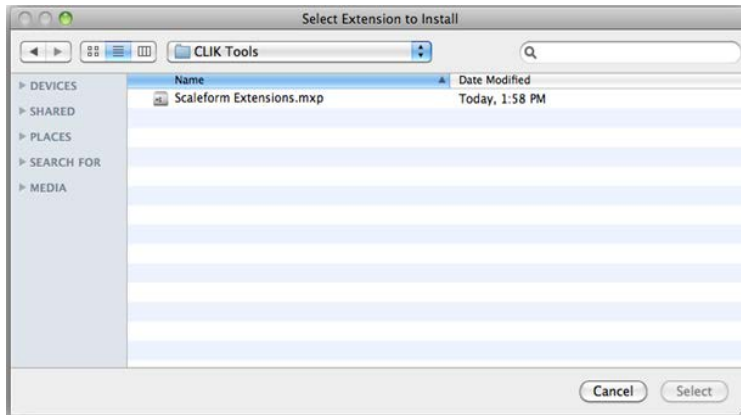
SWF

Scaleform™

1. Adobe Adobe CS3 , Windows
Windows® >Start , Photoshop® ,
>Programs>Adobe Extension Manager CS3 C:/Program Files/Adobe/Adobe
Extension Manager , Finder>Applications>Adobe Extension Manager
CS3>Extension Manager Adobe
2. Adobe CS3 ,Adobe Windows® Start
, Photoshop® , Install
3. Windows C:/Program Files/Scaleform/GFx SDK 4.3/Resources/Tools/
: scaleform_gfx_4.3_macos/Resources/Tools/
4. MXP :Scaleform Extensions.mxp.



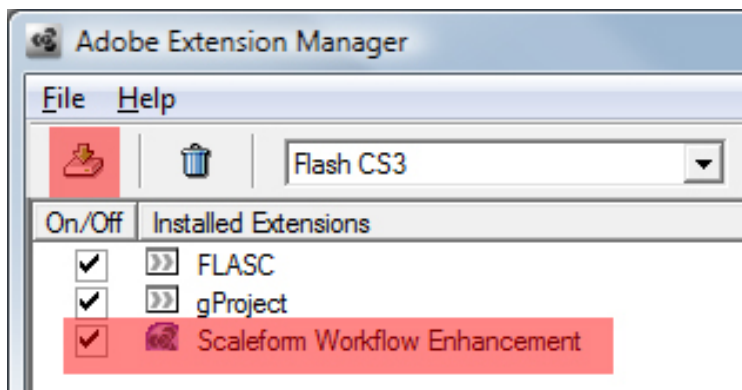
1: Scaleform Extensions.mxp Windows



2: Scaleform Extensions.mxp

5. Accept

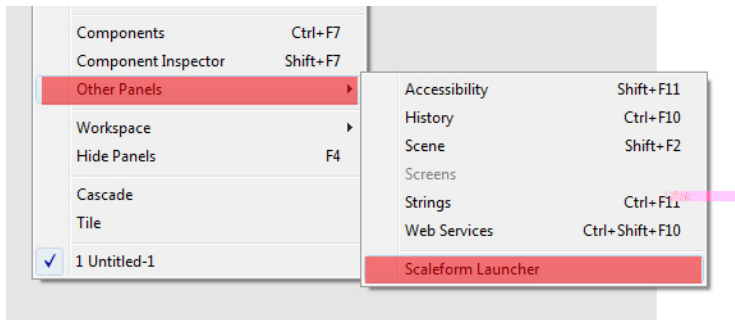
6. OK



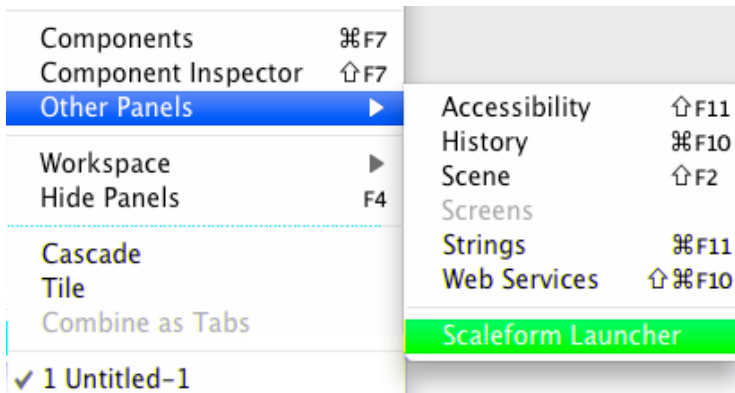
3: CS3 Adobe Extension Manager (Windows)

4: CS3 Adobe Extension Manager (Mac)

7. Adobe
8. Flash Flash (ActionScript 2.0 or 3.0)
9. Flash



5: Scaleform Launcher (windows)



6: Scaleform Launcher (Mac)

2.8 CS4, CS5 Scaleform

Scaleform

SWF

Scaleform™

1. CS4 or CS5 Flash *help* Adobe
Manage Extensions
:

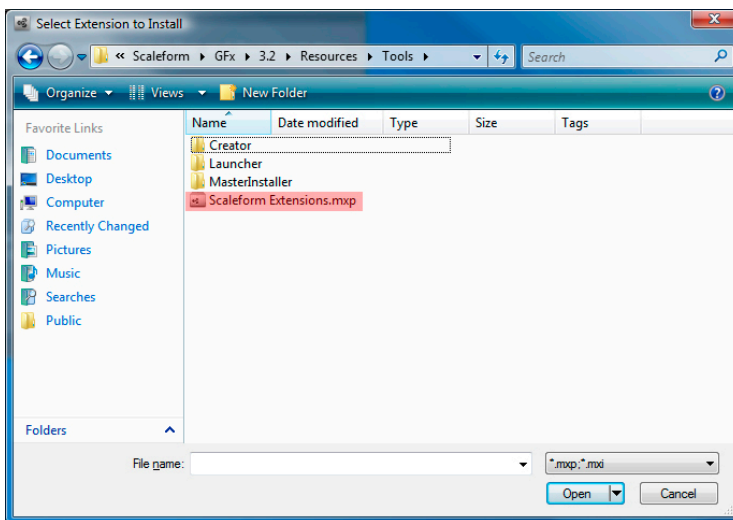
Flash

Adobe

2. *Install*

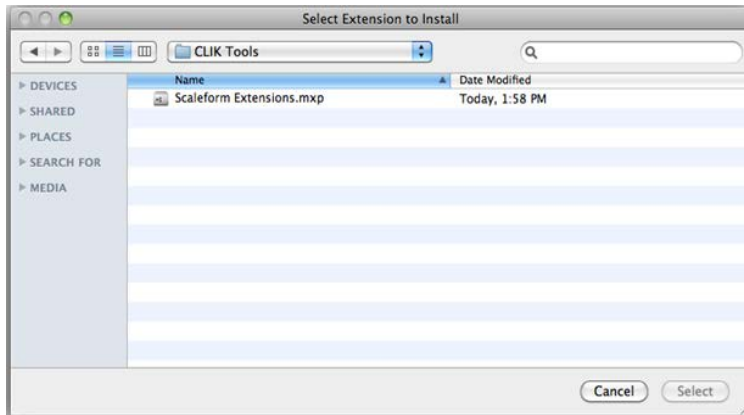
3. Windows : *C:/Program Files/Scaleform/GFx SDK 4.3/Resources/Tools/*
scaleform_gfx_4.3macos/Resources/Tools/

4. MXP *Scaleform Extensions.mxp.*



7: Scaleform Extensions.mxp

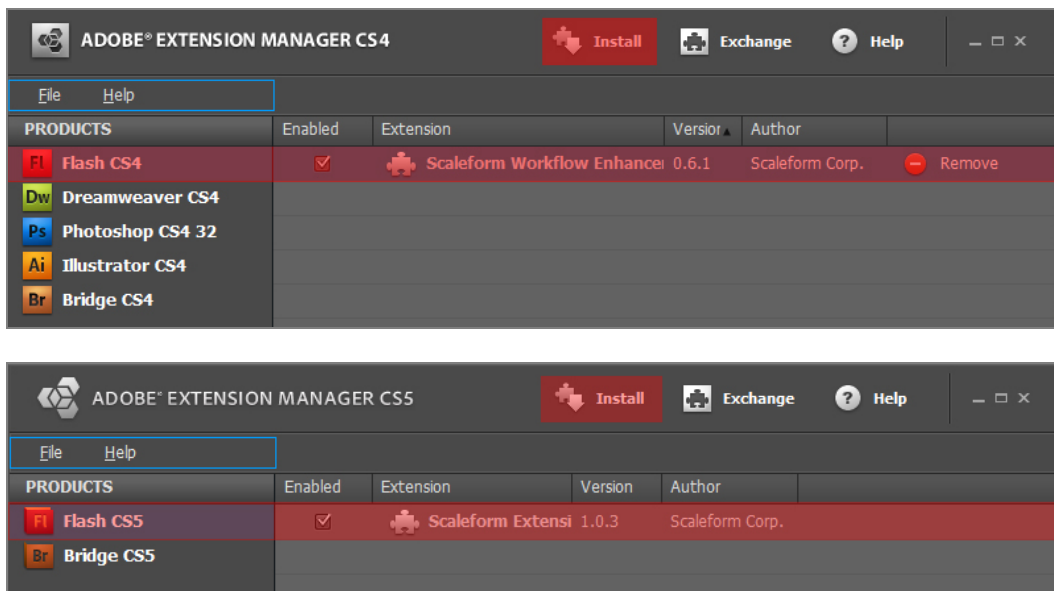
(Windows)



8: Scaleform Extensions.mxp (Mac)

5. *Accept*

6. Flash CS4 or CS5 OK



9: CS4 (top) and CS5 (bottom) Adobe Extension Manager

7. Adobe

8. Flash

9. Flash

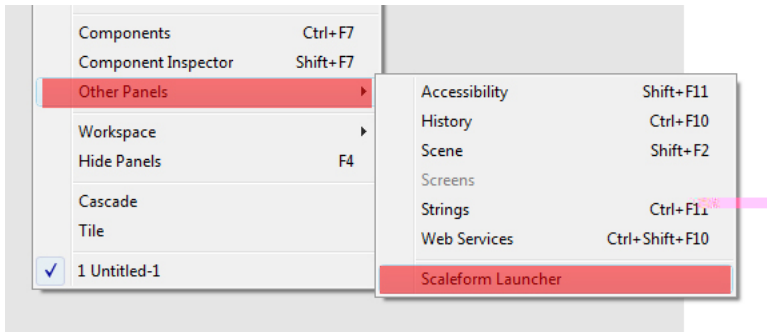


Figure 10: Scaleform Launcher panel (Windows).

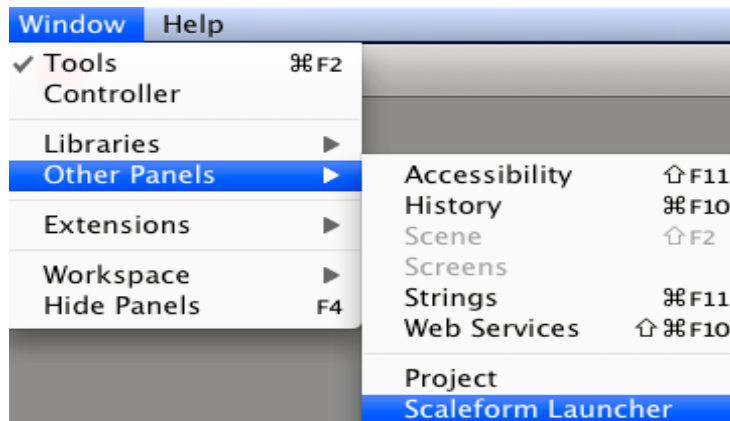
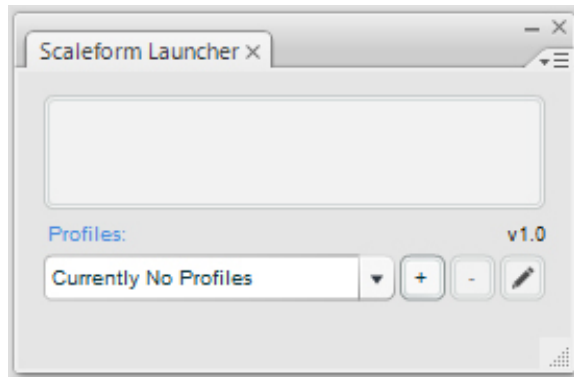


Figure 11: Scaleform Launcher panel (Mac).

2.9 *Scaleform*

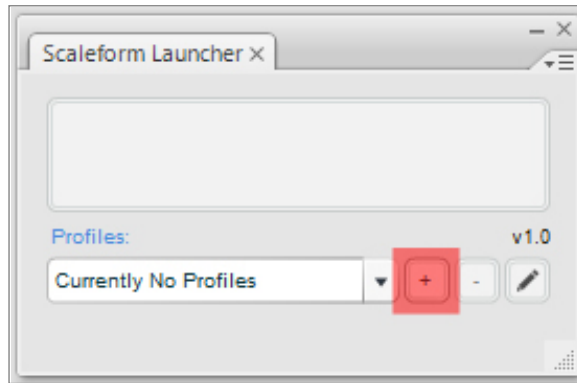
Scaleform *Scaleform* / SWF *Scaleform*
 Scaleform SWF Adobe CLIK
 Scaleform



12:Scaleform Launcher Panel

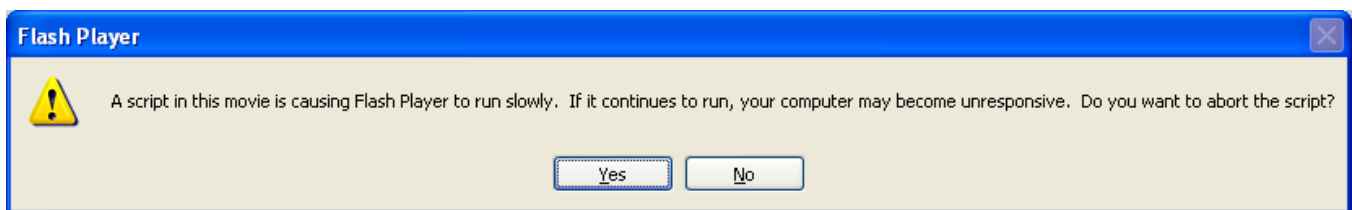
Add Profile

»



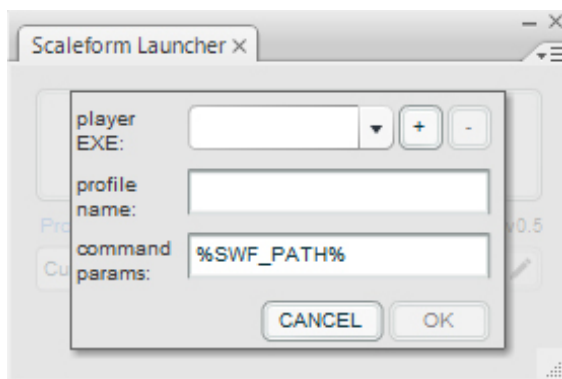
13: +

: Add Profile
 “the script is causing Flash to run slowly ” No



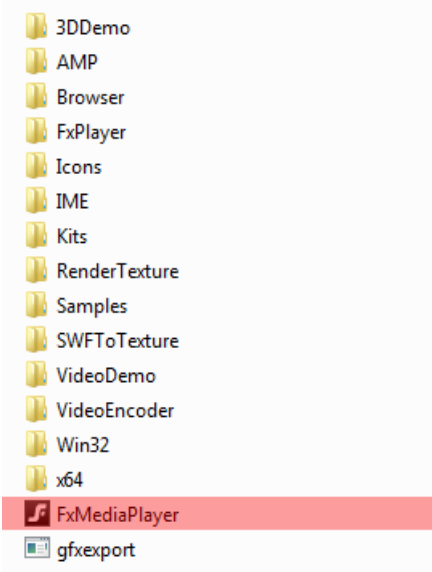
14: script

2. [+] Scaleform



15

3. *GfxMediaPlayer*
Scaleform *GfxMediaPlayer*
C:/Program Files/Scaleform/GFx SDK 4.3/Bin



16: **GfxMediaPlayer**

,
scaleform_gfx_4.3_macos/Bin/MacOS/FxPlayer_DebugOpt.app/Contents/MacOS
GfxPlayer_DebugOpt

17: **GfxPlayer_DebugOpt**

4. *Player Name*

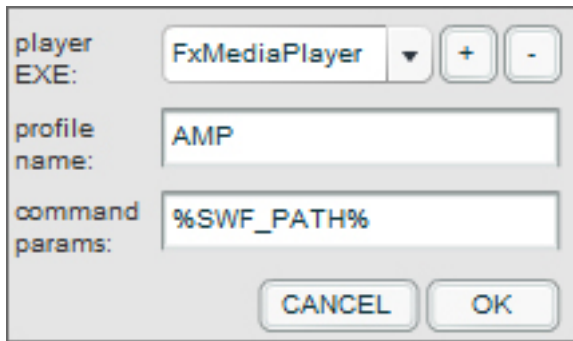
5. *OK*



```
6. Add Profile , profile name Windows
   , 'AMP' , 'DebugOpt'

7. Scaleform %SWF_PATH% '
: '%SWF_PATH% -nh '

SWF Scaleform HUD
```

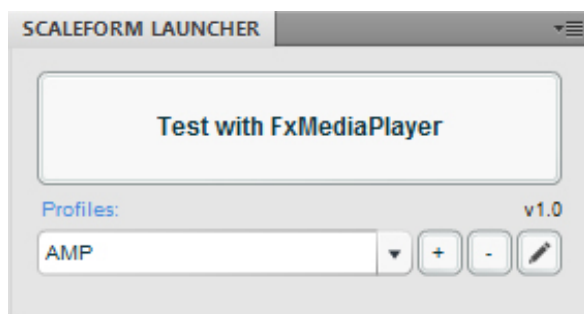



20: AMP

21: GfxPlayerDebugOpt

8. OK.

9. Scaleform Launcher Profiles new

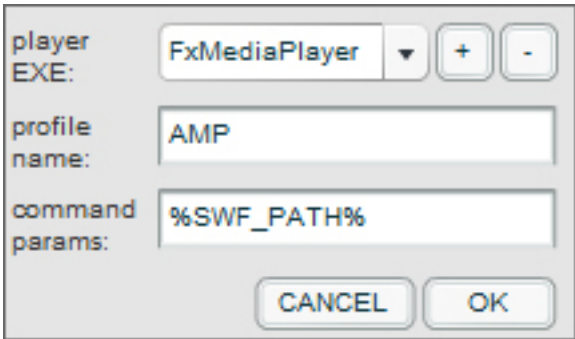


22: Scaleform Windows AMP

23: Scaleform 启动面板和苹果系统的 Debug_Opt 外貌

:

- 1. Scaleform Launcher Profiles
- 2. Edit
- 3. EXE
- 4. OK.



24: (Windows)

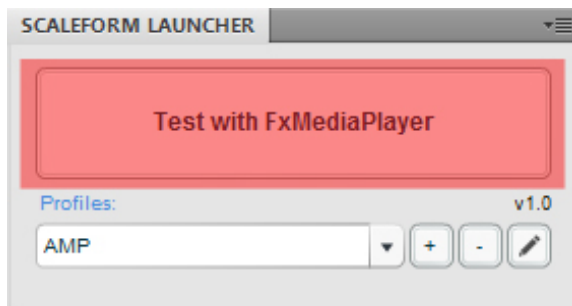
25: (Mac)

:

1. *Scaleform* Select
2. [-]
3. OK Cancel

SWF

1. SWF
2. *Scaleform* Profiles
3. Test with... [Executable Name]
4. Scaleform SWF



26: Test with GfxMediaPlayer SWF (Windows)

27: Test with GfxPlayer_DebugOpt SWF (Mac)

通過鍵盤快速鍵啟動播放機

- | | Flash | Scaleform Player |
|-----|---------------------------------|-------------------------|
| 1. | <i>Edit</i> | |
| 2. | <i>Keyboard Shortcuts</i> | |
| 3. | Adobe Standard " "(Current) : " | "(Keyboard Shortcuts) |
| 4. | Duplicate Set | |
| 5. | () | |
| 6. | OK | |
| 7. | Commands | "Drawing Menu Commands" |
| 8. | (+) | |
| 9. | Scaleform Player Publish() | |
| 10. | (+) | (Shortcut) |
| 11. | (Press key) | (Ctrl+Enter) |
| | Flash | Ctrl+Enter |
| 12. | Change | |
| 13. | , Reassign | |
| 14. | | |
| 15. | Test Movie | Flash |
| 16. | (+) | |
| 17. | (Press key) | (Ctrl+()) |
| 18. | Change | |
| 19. | OK " " | |

2.10

2.10.1

Scaleform 4.3 Eval *sf_license.txt*
sf_license.txt , -
3AAAA1BB23B8ZCCCC4CCEOF5JCZ08DDDDDEF
sf_license.txt
Scaleform 4.3 SDK *sf_license.txt*
sf_license.txt
Scaleform 4.3 *sf_license.txt*
The Scaleform trial period expired on 1(m) 1(d) 2008, please contact Scaleform for an extension. Error:
106
sf_license.txt
Failed to open the Scaleform license file, please make sure sf_license.txt exists in the working
directory. Error: 101

Windows / Mac® / Linux®

- 1.
2. GFXSDK
3. GFXSDK HOME

Xbox 360®

1. XDK - D:/\$ (SolutionName)/sf_license.txt

“PSP®”(PlayStation®Portable)”

5.
Linux - /usr/local/psp/FxPlayer/sf_license.txt
Windows - C:/usr/local/psp/FxPlayer/sf_license.txt

“PLAYSTATION®3”(PS3™)

1. /app_home/sf_license.txt
2. /host_root/usr/local/cell/FxPlayer/sf_license.txt
3. /dev_bdvd/PS3_GAME/USRDIR/sf_license.txt

PLAYSTATION®Vita”(PS Vita™)

1. sf_license.txt "FxPlayer" , ,

3DS™

1. Lib/3DS/target sf_license.txt

Wii™

1. dvddata/FxPlayer/sf_license.txt

iOS

1. sf_license.txt Xcode , Xcode (,Scaleform.app)

Android

1. sf_license.txt “/sdcard” (,)

2.11

Scaleform

Scaleform <http://gameware.autodesk.com/scaleform>

<http://gameware.autodesk.com/scaleform/developer/>

Scaleform

FAQ

Ticketing System

Scaleform Ticketing System

Scaleform

Scaleform

Scaleform

Scaleform

2.12

Scaleform

gameware@autodesk.com

Scaleform

3

Scaleform 4.3

Scaleform 4.3

Scaleform 4.3

Scaleform 3.0

Scaleform 4.3

Flash AS 2.0

Scaleform 4.3

Doc

C:/Program

Files/Scaleform/GFx SDK 4.3/Doc/

1. SDK
2. Windows : Start->Programs->Scaleform->GfX SDK 4.3->Documentation

3.1 Flash

Flash AS2

Flash

Scaleform

IGDA Flash

SIG

http://www.igda.org/wiki/Flash_SIG

Flash

IGDA SIG

Flash `ActionScript`

- [Lynda.com](#)
- [Essential ActionScript 2.0 by Collin Moock](#)
- [ActionScript: The Definitive Guide by Collin Moock](#)

3.2

Scaleform 4.3

Scaleform 4.3

- *CLIK* CLIK CLIK Illustrator® Photoshop®
- *CLIK* CLIK CLIK CLIK *CLIK*
- USM After Effects® Adobe Premiere
Scaleform 4.3 Flash
Scaleform
- *IME*

3.3

Scaleform 4.3 Flash

Scaleform 4.3

- DirectX 9 Scaleform 4.3 3D
- CHM Scaleform 4.3
- *DrawText API* Scaleform 4.3 DrawText API C++
ActionScript
- *ActionScript* Scaleform 4.3 ActionScript
- Scaleform 4.3
Scaleform C++ API

- *Scale9Grid*: Scaleform 4.3 Scale9Grid
- *XML Overview*: Scaleform 4.3 XML
- Flash Excel (XLS) Scaleform 4.3 Flash
- Scaleform 4.3 AMP ActionScript
- *IME* : Scaleform 4.3 IME IME
-

3.4

UE3
SDK gameware@autodesk.com

- *IME Xbox 360 Configuration*: Xbox 360 IME
- *IME PS3* : PS3 IME
- *Unreal Engine 3* : Flash 3
- *Unreal Engine 3* : Scaleform 4.3 UE3
Scaleform 4.3 UE3 Kismet UnrealScript Unreal Editor
- README
SDK ()

3.5

Scaleform 4.3

Scaleform

“ ”

Web

<http://gameware.autodesk.com/scaleform/developer/?action=demo>

Flash

GfX 4

- *Scaleform* Scaleform

Scaleform SDK

: C:/Program Files (x86)/Scaleform/GfX SDK

4.3/Apps/Samples/GfXPlayerTiny

- *Scaleform* SWF Flash 3D

(PC): C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc90/

GfXPlayerSWFToTexture/SWFToTexture_D3D9_Release_Static.exe

: C:/Program Files (x86)/Scaleform/GfX SDK 4.3/Apps/Samples/RenderTexture

- *Scaleform* SWF Flash 3D Flash

(PC):: C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc90/

GfXPlayerTextureInSWF/TextureInSWF_D3D9_Release_Static.exe

: C:/Program Files (x86)/Scaleform/GfX SDK 4.3/Apps/Samples/RenderTexture

Linux ,

Scaleform SDK

Bin/<platform>/

3.6

Scaleform 4.3 SDK

Scaleform

<http://gameware.autodesk.com/scaleform/developer/?action=doc>

Scaleform SDK
Started) "(Video Tutorials) "(Getting
<http://gameware.autodesk.com/scaleform/developer/?action=vid>

4

Scaleform 4.3

Scaleform

Scaleform 4.3

Flash

Scaleform 4.3

Scaleform 4.3

Scaleform 4.3

UI

/

UI

Flash

Flash

FAQ

Scaleform :

<http://gameware.autodesk.com/developer/forums/>

4.1

4.1.1 UI

Scaleform

Scaleform 4.3

2D Flash

Flash

ActionScript

Flash

4.1.2 UI

Flash AS2
CLIK

4.1.3 UI

CS3 CS4 Flash UI CLIK

Scaleform 4.3 Flash

4.2

Scaleform 4.3 Window Scaleform 4.3 Flash
SWF SWF GFX Scaleform 4.3

, Windows , Scaleform 4.3 SDK "Scaleform SDK Browser"
SDK Browser " "(Programs) Scaleform

Start->Programs->Scaleform ->GfX SDK 4.3->GfX Players

Windows

C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc80/GfXPlayer

C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc90/GfXPlayer

, SDK Browser Scaleform GfX Player

Windows Windows

C:/Program Files/Scaleform/GfX SDK

4.3/Bin/Data/AS2/Samples/3DDemo/

3DDemo	SWF			
, (Windows)	Linux, Finder()	Scaleform SDK 4	Bin/<platform>/	
SWF	, Scaleform SDK 4.3	Bin/3DDemo	SWF	SWF
SWF				SWF

:

Linux - Bin/i386-linux/FxPlayer <swf >

MacOS - Bin/MacOs/FxPlayer.app/Contents/MacOS/FxPlayer <swf >

Flash

Scaleform SDK 4 /Bin/Data/AS2/Samples

4.3

Scaleform 4.3	AS 2.0, 3.0	Flash CS3, CS4	CS5
		ActionScript 2 or ActionScript 3	
	Flash		File->New->Flash File (ActionScript 2.0 or
ActionScript 3.0)	◦	Flash	Flash Player 10
	Scaleform 4.3	AS 2.0	
sf_4.3_flash_support.pdf		AS 2.0	